

STATE TECHNICAL UNIVERSITY

END SEMESTER EXAMINATION 2025-2026 | ANSWER BOOKLET

CANDIDATE ROLL NO: 2026-CS-PH-027
CANDIDATE NAME: Khushi Dutta
EXAMINATION / COURSE: B.Tech Semester II
SUBJECT & CODE: Physics (CS 201)
BOOKLET SERIAL NO: BK-PH-80027

BARCODE SCAN TRACKING



BK-PH-80027

Q.No	Q1	Q2	Q3	Q4	Q5
Max	14	14	14	14	14
Marks					

START WRITING YOUR ANSWERS BELOW. DO NOT WRITE IN THE MARGINS.

Q.1

Ans 1. De Broglie Waves - Group Velocity:

1. Matter waves have wavelength $\lambda = h / mv$.
2. Phase velocity v_p is the velocity of individual crests: $v_p = \omega/k = E/p$.
3. Group velocity v_g is velocity of packet = $d(\omega)/dk$.
4. Relation: For kinetic energy $E = p^2/(2m)$, $v_g = dE/dp = p/m = v$.

Thus group velocity is equal to the particle velocity. This confirms that the particle moves with the center of the wave packet.

Q.2

Ans 2. Interference of Light:

Interference occurs when two coherent waves meet.

- Conditions: Coherent sources, monochromatic light, narrow slit sources.
- Division of Wavefront: Splits wavefront using slits (e.g. Young's Double Slit).
- Division of Amplitude: Splits amplitude by partial reflection (e.g. Newton's Rings, Thin films).
- In Thin films, path difference $\Delta = 2 \mu t \cos(r) - \lambda/2$.
- For constructive interference: $2 \mu t \cos(r) = (2n-1) \lambda / 2$.
- For destructive interference: $2 \mu t \cos(r) = n \lambda$.

Q.3

Ans 3. Lasers & Optical Fibres:

1. Laser Basics:

- Stimulated emission produces coherent photons.
- Population inversion is necessary where excited atoms exceed ground atoms ($N_2 > N_1$).
- Main components: Active medium, Pumping source, Optical cavity.

2. Optical Fibres:

- Work on Total Internal Reflection (TIR).
- Core refractive index n_1 is higher than Cladding n_2 .
- Numerical Aperture $NA = \sqrt{n_1^2 - n_2^2}$.
- Acceptance angle $\theta_a = \arcsin(NA)$.

Q.4

Ans 4. Maxwell's Equations:

Maxwell unified electricity and magnetism into four equations:

1. $\text{div } \mathbf{D} = \rho$ (Electric charges produce electric flux)
2. $\text{div } \mathbf{B} = 0$ (No magnetic monopoles)
3. $\text{curl } \mathbf{E} = - \frac{d\mathbf{B}}{dt}$ (Faraday's Law)
4. $\text{curl } \mathbf{H} = \mathbf{J} + \frac{d\mathbf{D}}{dt}$ (Ampere-Maxwell Law with displacement current $\mathbf{J}_d = \frac{d\mathbf{D}}{dt}$)

Boundary conditions:

- Tangential $\mathbf{E}$ field is continuous across interface: $\mathbf{E}_{1t} = \mathbf{E}_{2t}$.
- Normal $\mathbf{B}$ field is continuous: $\mathbf{B}_{1n} = \mathbf{B}_{2n}$.
- Normal $\mathbf{D}$ field: $\mathbf{D}_{1n} - \mathbf{D}_{2n} = \text{surface charge density}$.
- Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ represents power density.

Q.5

Ans 5. Solid State Physics:

1. Crystal Structures:

- SC: 1 atom/cell, packing fraction = 52%.
- BCC: 2 atoms/cell, packing fraction = 68%.
- FCC: 4 atoms/cell, packing fraction = 74%.
- Miller indices (hkl) denote crystal planes.

2. Semiconductors:

- Intrinsic: Pure silicon with equal electrons and holes ($n = p$).
- Extrinsic: Doped with impurities (n-type with donors, p-type with acceptors).
- Fermi level E_f is the highest filled energy level at 0 Kelvin.
- Direct bandgap emits photons (GaAs), indirect emits phonons/heat (Si).

--- End of Answer Booklet (All 5 questions attempted) ---

Candidate Sign: Khushi